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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

2 - Proofs in Propositional Logic

All deductions must use the DN natural deduction system and the Fitch notation given in class.

Notes for obtaining a deduction from a formula φ

- If the formula φ is an implication $\psi \to \theta$, assume ψ and deduce θ; then apply the rule of introduction of implication.
- If the formula φ is a negation $\neg\psi$, assume ψ and deduce F; then apply the rule of introduction of negation.
- By contradiction: assume $\neg\varphi$ and deduct F. In case there is a negation $\neg\psi$ in the premises, one can deduce ψ to get F.
- Suppose one of the premises is $\varphi \vee \psi$ and one wants to deduce γ. We can: assume φ and deduce γ, assume ψ and deduce γ; then apply the rule of elimination of the disjunction.
- Suppose an assumption is $\neg\varphi$ and we want to get $\mathbf{F}$: we can try to get φ and apply the intro rule of $\mathbf{F}$.
- Suppose we have an implication $\phi \to \theta$. We can then try to deduct ϕ and apply the rule of elimination of implication (modus ponens) to get θ.

Note that each step of the deduction must be obtained by applying a rule or be a repetition of a formula already deduced or a premise. Thus each deduction step must either be an assumption or result from the application of a rule to formulas already deduced, so "semantic equivalences" cannot be used.

Note: In case you use derived rules, you should prove then separately.

2.1. Match each of the following tautologies to one of the logical arguments.

Tautologies

(1) $p \vee \neg p$
(2) $p \wedge q \to p$
(3) $p \to p \vee q$
(4) $(p \to q) \to (\neg q \to \neg p)$
(5) $((p \to q) \wedge (\neg p \to q)) \to q$
(6) $(((p \vee q) \to r) \wedge p) \to r$

Arguments

(a) If x is a real number such that $0 \leq x \leq 1$, then $x \leq 1$.
(b) Let n be a natural number such that n is divisible by 3. Then n is divisible by 2 or n is divisible by 3.
(c) John gets frequency if he does his work in C or Java. John did the work in C. So John obtained frequency.
(d) Let x be a real number; we have $x < 0$ or $x \geq 0$.
(e) Let n be an integer; if n is divisible by 4 then n is divisible by 2. Then, if n is not divisible by 2, then is not divisible by 4.
(f) Let n be an integer; if n is even, then $n^2 - n = (n-1)n$ is even; if n is odd, then $n^2 - n = (n-1)n$ is even. Therefore $n^2 - n$ is always even.

2.2. Indicate whether the following deductions are valid. If so, indicate the rules used in each step. If not valid, indicate which step it is not correct and if there is a correct deduction with the same premises and conclusion.

a)

1	$p \vee q$
2	| p
3	| p
4	| q
5	| q
6	$p \wedge q$

b)

1	$(p \wedge q) \vee r$
2	| $p \wedge q$
3	| p
4	| $p \vee r$
5	| r
6	| $p \vee r$
7	$p \wedge r$

c)

1	$\neg(p \vee \neg q)$
2	| $\neg q$
3	| $p \vee \neg q$
4	| **F**
5	$\neg\neg q$
6	q

d)

1	$p \to q$
2	| p
3	| q
4	| $\neg p \vee q$
5	$\neg p \vee q$

e)

1	$p \to \neg p$
2	| p
3	| $\neg p$
4	| **F**
5	$\neg p$

f)

1	$\neg p \to \neg q$
2	| q
3	| $\neg q$
4	| **F**
5	| p
6	$q \to p$

2.3. Using rules of logical inference, build a proof for the conclusion "It rained" from the hypothesis "If it doesn't rain or if there is no fog, then the sailing will take place and the rescue demonstration will be carried out", "If the sailing competition takes place then the trophy will be awarded" and "The trophy was not awarded".

2.4. Show that:

(a) $(p \wedge q) \to r, p \vdash q \to r$

(b) $q \to p, q \to \neg p \vdash \neg q$

(c) $(\neg p \vee \neg q) \vee r, q \vee r, p \vdash r$

(d) $p \vee \neg q, \neg r \to q, r \to \neg s, s \vdash p$

(e) $p \to (q \vee r), q \to s, r \to t \vdash p \to (s \vee t)$.

(f) $p \to (q \wedge r), q \vee s \to t, p \vee s \vdash t$.

(g) $(t \to (r \vee p)) \to ((\neg r \vee k) \wedge \neg k) \vdash \neg r$.

2.5. Show that:

(a) $(\delta \wedge \neg\varphi) \to \psi, \neg\psi, \delta \vdash \varphi$

(b) $\vdash (\delta \to \psi) \to ((\neg\delta \to \neg\varphi) \to (\varphi \to \psi))$

(c) $\vdash (\delta \wedge \varphi) \to \varphi$

(d) $\vdash \delta \to (\neg\varphi \to \neg(\delta \to \varphi))$

(e) $\vdash ((\varphi \to \delta) \wedge (\delta \to \varphi)) \to ((\varphi \vee \delta) \to (\varphi \wedge \delta))$

2.6. Indicate which of the following deductions are valid and which are not:

(a) $\neg\delta, \delta \vee \varphi \vdash \varphi$

(b) $\varphi \wedge \neg\varphi \vdash \neg(\delta \to \psi) \wedge (\delta \to \psi)$

(c) $\neg(\neg\delta \vee \varphi) \vdash \delta$

(d) $\delta \vee \psi, \neg\psi \vee \varphi \vdash \delta \vee \varphi$

(e) $\vdash (\neg\delta \to \delta) \to \delta$

(f) $\neg\delta \to \neg\psi \vdash \neg\psi \to \neg\delta$

2.7. Justify the validity or falsity of each of the following statements with a proof or a counter-example:

(a) if $\vdash \psi$, then $\delta_1, \ldots, \delta_n \vdash \psi$;

(b) if $\delta_1, \ldots, \delta_n \vdash \psi$, then $\vdash \psi$;

(c) if $\delta_1, \delta_2 \vdash \psi$, then $\delta_2, \delta_1 \vdash \psi$;

(d) if $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1, \delta_2 \vdash \varphi$, then $\delta_1, \delta_2 \vdash \psi$;

(e) if $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1 \vdash \varphi$, then $\delta_1, \delta_2 \vdash \psi$;

(f) if $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1 \vdash \varphi$, then $\delta_1 \vdash \psi$;

(g) if $\delta_1 \vdash \varphi \to \psi$ and $\delta_2 \vdash \varphi$, then $\delta_1, \delta_2 \vdash \psi$;

(h) $\delta, \delta \vdash \psi$ if and only if $\delta \vdash \psi$.

2.8. Show, without using the completeness of DN, the following deductive equivalences. Note that you have to show that the second formula is deduced from the first formula and vice versa.

(a) $\neg\delta_1 \wedge \delta_2 \vdash \delta_1 \vee \delta_2$

(b) $\delta_1 \wedge \delta_2 \vdash \delta_1 \vee \delta_2$

(c) $(\psi \wedge \varphi) \wedge \delta \dashv\vdash \psi \wedge (\varphi \wedge \delta)$

(d) $\neg(\neg\delta_1 \wedge \neg\delta_2) \dashv\vdash \delta_1 \vee \delta_2$

(e) $\delta_1 \to \delta_2 \dashv\vdash \neg(\delta_1 \wedge \neg\delta_2)$

(f) $\delta_1 \wedge \delta_2 \dashv\vdash \neg(\neg\delta_1 \vee \neg\delta_2)$

(g) $(\varphi \vee \psi) \to \delta \dashv\vdash (\varphi \to \delta) \wedge (\psi \to \delta)$

(h) $(\delta \vee \varphi) \wedge (\delta \vee \psi) \dashv\vdash \delta \vee (\varphi \wedge \psi)$

(i) $\varphi \to (\psi \wedge \delta) \dashv\vdash (\varphi \to \psi) \wedge (\varphi \to \delta)$

(j) $\delta_1 \wedge \neg\delta_2 \vdash \neg(\delta_1 \to \delta_2)$

2.9. Let Σ and Γ be sets of propositional logic formulas such that for any formula δ one has $\Sigma \vdash \delta$ or $\Gamma \vdash \delta$.

(a) Show that $\Sigma \cup \Gamma$ is not satisfiable.

(b) Show that one of the sets Σ or Γ is not satisfiable.

Natural Deduction (DN) for propositional logic (Fitch notation)

	Introduction	**Elimination**
$\wedge$	φ $\vdots$ ψ $\vdots$ $\varphi \wedge \psi$ $\wedge$I	$\vdots$ $\varphi \wedge \psi$ $\vdots$ φ $\wedge$E $\vdots$ $\varphi \wedge \psi$ $\vdots$ ψ $\wedge$E
$\vee$	$\vdots$ φ $\vdots$ $\varphi \vee \psi$ $\vee$I $\vdots$ φ $\vdots$ $\psi \vee \varphi$ $\vee$I	$\varphi \vee \psi$ $\vdots$ φ / $\vdots$ / γ ψ / $\vdots$ / γ γ $\vee$E
$\neg$	φ / $\vdots$ / **F** $\neg\varphi$ $\neg$I	$\vdots$ $\neg\neg\varphi$ $\vdots$ φ $\neg$E
F	φ $\vdots$ $\neg\varphi$ $\vdots$ **F** **FI**	$\vdots$ **F** $\vdots$ φ **FE**
$\rightarrow$	φ / $\vdots$ / ψ $\varphi \rightarrow \psi$ $\rightarrow$I	φ $\vdots$ $\varphi \rightarrow \psi$ $\vdots$ ψ $\rightarrow$E